

Quiz 3

Instructions: There are totally 4 problems. You must show all of your work to receive a credit for a correct answer.

Problem 1. (5 points) Find all first and second-order partial derivatives of $f(x, y) = xe^{x^2-y^2}$.

$$\frac{\partial f}{\partial x} = e^{x^2-y^2} + xe^{x^2-y^2}(2x) = (1+2x^2)e^{x^2-y^2}$$

$$\frac{\partial f}{\partial y} = xe^{x^2-y^2}(-2y) = -2xye^{x^2-y^2}$$

$$\frac{\partial^2 f}{\partial x^2} = (4x)e^{x^2-y^2} + (1+2x^2)e^{x^2-y^2}(2x) = (6x+4x^3)e^{x^2-y^2}$$

$$\frac{\partial^2 f}{\partial y^2} = (-2x)e^{x^2-y^2} + (-2xy)e^{x^2-y^2}(-2y) = (-2x+4xy^2)e^{x^2-y^2}$$

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x} = (1+2x^2)e^{x^2-y^2}(-2y) = (-2y-4x^2y)e^{x^2-y^2}$$

Problem 2. (5 points) Use the **Chain Rule** to find $D(f \circ g)(s, t)$ when $f(x, y) = e^{x+2y}$

and $g(s, t) = \left(\frac{s}{t}, \frac{t}{s}\right)$.

$$Df = [e^{x+2y} \quad 2e^{x+2y}]$$

$$Dg = \begin{bmatrix} \frac{1}{t} & -\frac{s}{t^2} \\ -\frac{t}{s^2} & \frac{1}{s} \end{bmatrix}$$

$$x = \frac{s}{t} \text{ and } y = \frac{t}{s}$$

$$\Rightarrow D(f \circ g)(s, t) = Df Dg = [e^{x+2y} \quad 2e^{x+2y}] \begin{bmatrix} \frac{1}{t} & -\frac{s}{t^2} \\ -\frac{t}{s^2} & \frac{1}{s} \end{bmatrix}$$

$$= [e^{x+2y} \left(\frac{1}{t} - \frac{2t}{s^2}\right) \quad e^{x+2y} \left(-\frac{s}{t^2} + \frac{2}{s}\right)]$$

$$= [e^{\frac{s}{t} + \frac{2t}{s}} \left(\frac{1}{t} - \frac{2t}{s^2}\right) \quad e^{\frac{s}{t} + \frac{2t}{s}} \left(-\frac{s}{t^2} + \frac{2}{s}\right)]$$

Problem 3. (5 points) Given a function $f(x, y, z) = \sin(xy) + z^2$. Find the directional derivative of f at the point $P(1, 0, 1)$ along the direction of the vector $\mathbf{v} = (1, 2, 2)$. In what direction does f increase the most rapidly at P and what is the corresponding rate of change?

$$\nabla f(x, y, z) = (\cos(xy), x \sin(xy), 2z) \Rightarrow \nabla f(1, 0, 1) = (0, 1, 2)$$

$$\mathbf{v} = (1, 2, 2) \Rightarrow \mathbf{u} = \frac{\mathbf{v}}{\|\mathbf{v}\|} = \frac{(1, 2, 2)}{\sqrt{1^2 + 2^2 + 2^2}} = \frac{1}{3}(1, 2, 2) = \left(\frac{1}{3}, \frac{2}{3}, \frac{2}{3}\right)$$

Thus we have

$$D_{\mathbf{u}} f|_{(1, 0, 1)} = \nabla f(1, 0, 1) \cdot \mathbf{u} = (0, 1, 2) \cdot \left(\frac{1}{3}, \frac{2}{3}, \frac{2}{3}\right)$$

$$= 0 + \frac{2}{3} + \frac{4}{3} = 2$$

f increase the most rapidly at $(1, 0, 1)$ in the direction of $\nabla f(1, 0, 1) = (0, 1, 2)$ and the corresponding rate of change is $\|\nabla f(1, 0, 1)\| = \sqrt{0^2 + 1^2 + 2^2} = \sqrt{5}$.

Problem 4. (5 points) Find an equation for the plane tangent to the surface defined by $x^2 - 2y^2 + 5xz = 7$ at the point $(-1, 0, -\frac{6}{5})$.

$$f(x, y, z) = x^2 - 2y^2 + 5xz$$

$$\Rightarrow \nabla f(x, y, z) = (2x + 5z, -4y, 5x)$$

thus a normal vector the surface at $(-1, 0, -\frac{6}{5})$ is

$$\nabla f(-1, 0, -\frac{6}{5}) = (2(-1) + 5(-\frac{6}{5}), -4 \cdot 0, 5(-1))$$

$$= (-8, 0, -5)$$

Thus an equation for the tangent plane is given by

$$(-8, 0, -5) \cdot (x - (-1), y - 0, z - (-\frac{6}{5})) = 0$$

$$-8(x+1) + 0 \cdot y - 5 \cdot (z + \frac{6}{5}) = 0$$

$$-8x - 5z = 14$$